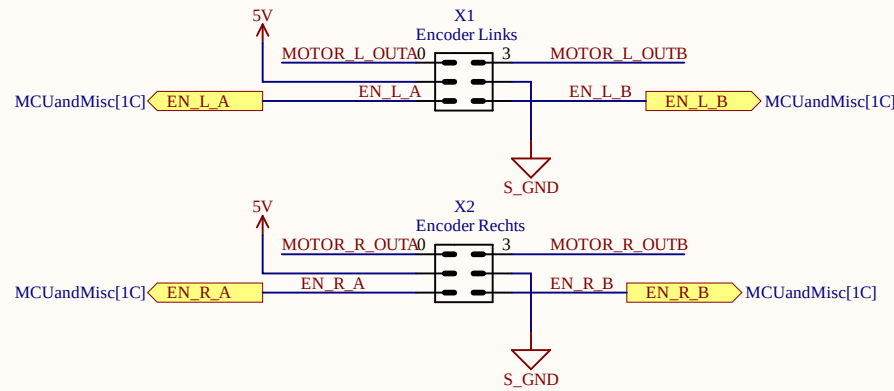


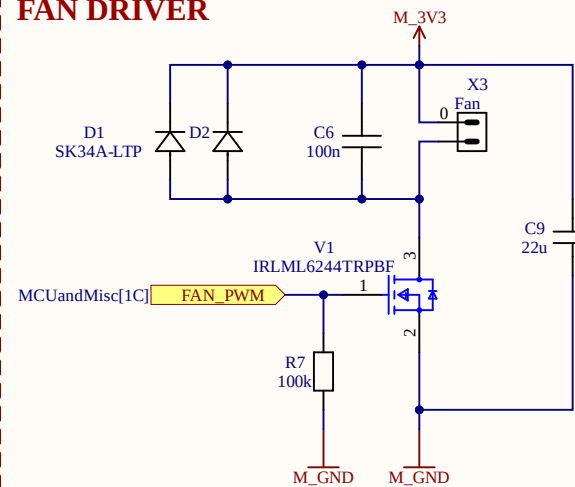
ENCODER and MOTOR CONNECTORS

The diagram illustrates the wiring for two encoders, X1 and X2, connected to an MCU. Both encoders are powered by a 5V supply and ground (S_GND). The motor outputs (MOTOR_L_OUTA0, MOTOR_L_OUTB for X1 and MOTOR_R_OUTA0, MOTOR_R_OUTB for X2) are connected to the MCU's EN_L_A, EN_L_B (for X1) or EN_R_A, EN_R_B (for X2) pins. The encoder links (Encoder Links for X1 and Encoder Rechts for X2) are connected to the MCU's EN_L_A, EN_L_B (for X1) or EN_R_A, EN_R_B (for X2) pins. The MCU is labeled MCUandMisc[1C].

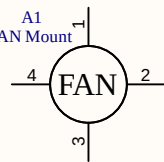


FAN DRIVER

The diagram illustrates a fan driver circuit. It features a motor (M3) connected to a 3V3 supply. A diode bridge (D1, D2) is connected to the motor. A capacitor (C6, 100n) is connected across the motor terminals. A fan (X3) is connected to the motor. A MOSFET (V1, IRLML6244TRPBF) is driven by a PWM signal (FAN_PWM) from the MCU and Misc[1C]. A resistor (R7, 100k) is connected between the gate and source of the MOSFET. A capacitor (C9, 22u) is connected across the motor terminals. The motor is connected to M_GND.



A diagram of a FAN (Fan Assembly) with a central circle labeled "FAN". Four lines extend from the circle, numbered 1 through 4. Line 1 points upwards and is labeled "A1 FAN Mount". Line 2 points to the right. Line 3 points downwards. Line 4 points to the left.

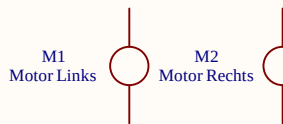


MOTOREN



M1 Motor Links

M2 Motor Rechts



POWER

The schematic diagram illustrates the power management section of the TMS320C6748 evaluation module. It features two main integrated circuits: the TPS62932DRLR (N3) and the LP38690SD-3.3/NOPB (N4).

TPS62932DRLR (N3) Configuration:

- VIN:** Connected to VBat through a network of capacitors (C16, C17, C18, C24) and a resistor (R13).
- EN:** Connected to VBat.
- SS:** Connected to GND through a capacitor (C24).
- SW:** Connected to the output of the converter through an inductor (L1) and a capacitor (C15).
- FB:** Connected to the output of the converter through a resistor (R9).
- RT:** Connected to GND through a resistor (R13).
- GND:** Connected to GND.

LP38690SD-3.3/NOPB (N4) Configuration:

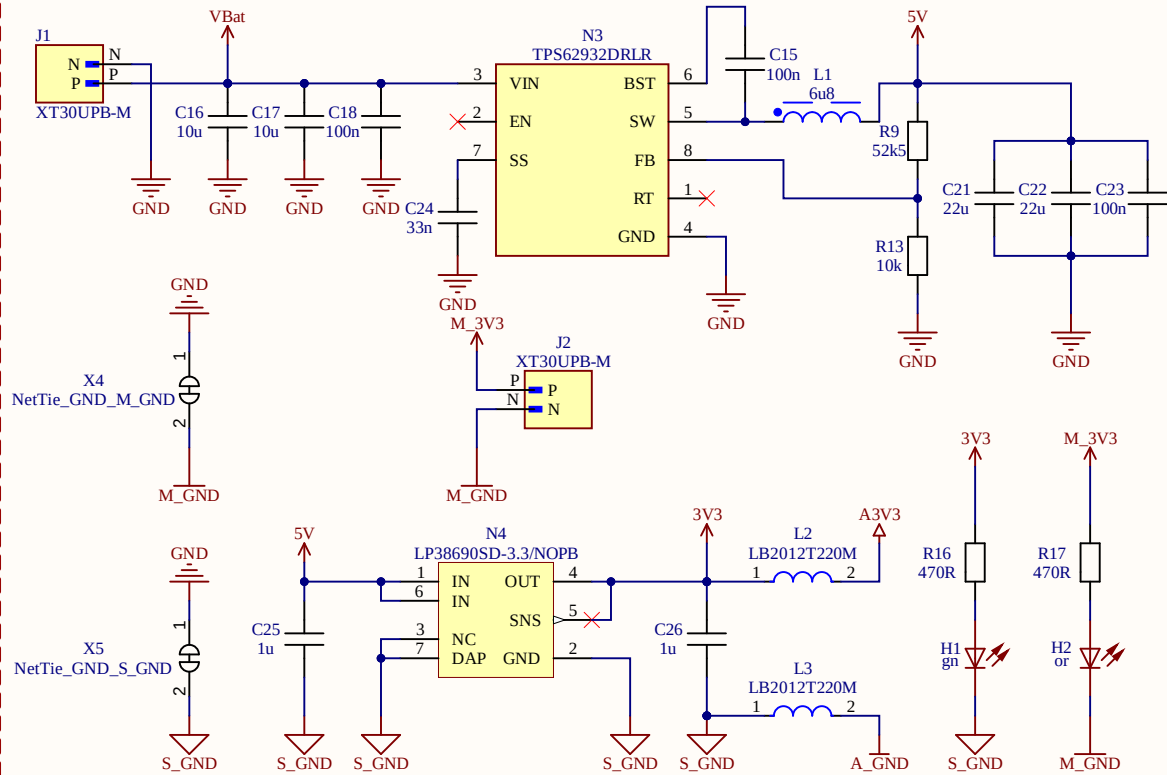
- IN:** Connected to 5V through a capacitor (C25).
- IN IN:** Connected to 5V.
- NC:** Connected to GND.
- DAP:** Connected to GND.
- OUT:** Connected to the output of the LDO through a capacitor (C26).
- SNS:** Connected to the output of the LDO through a resistor (R16).
- GND:** Connected to GND.

Output and Connections:

- The output of the LDO is connected to the 3V3 pin, which is also connected to the M_3V3 pin.
- The output of the DC-DC converter is connected to the 5V pin, which is also connected to the M_5V pin.
- The output of the DC-DC converter is also connected to the 3V3 pin through a resistor (R17).
- The output of the DC-DC converter is also connected to the M_3V3 pin through a resistor (R17).

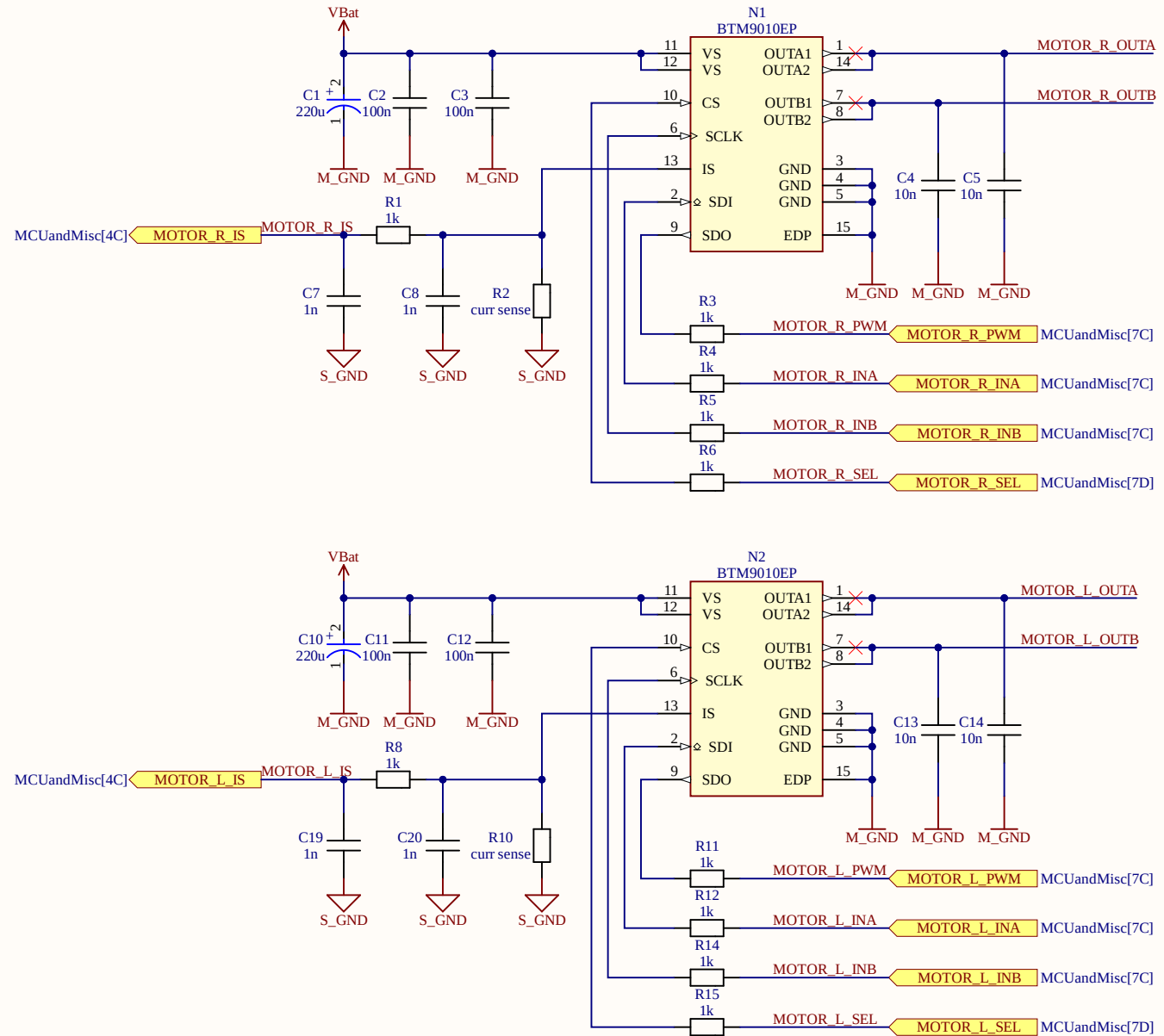
Other Components:

- XT30UPB-M (J1, J2):** Connectors for the power supply and ground.
- NetTie_GND_M_GND (X4):** A net tie between GND and M_GND.
- NetTie_GND_S_GND (X5):** A net tie between GND and S_GND.



MOTOR DRIVERS

The diagram illustrates two motor driver circuits, N1 and N2, both utilizing the BTM9010EP IC. Each circuit is powered by VBat and includes decoupling capacitors (C1, C2, C3 for N1; C10, C11, C12 for N2) and a current sense resistor (R2 for N1; R10 for N2) connected to S_GND. The control signals for both drivers are provided by the MCU via various pins, including MOTOR_R_IS, MOTOR_L_IS, MOTOR_R_PWM, MOTOR_L_PWM, MOTOR_R_INA, MOTOR_L_INA, MOTOR_R_INB, MOTOR_L_INB, MOTOR_R_SEL, and MOTOR_L_SEL. The output signals for the right motor (N1) are MOTOR_R_OUTA and MOTOR_R_OUTB, while the output signals for the left motor (N2) are MOTOR_L_OUTA and MOTOR_L_OUTB. The BTM9010EP IC pins are labeled with their functions: VS, CS, SCLK, IS, SDI, SDO, EDP, OUTA1, OUTA2, OUTB1, and OUTB2. The pins are connected to the MCU signals as follows: VS to VBat, CS to MOTOR_R_IS (N1) or MOTOR_L_IS (N2), SCLK to MOTOR_R_IS (N1) or MOTOR_L_IS (N2), IS to MOTOR_R_PWM (N1) or MOTOR_L_PWM (N2), SDI to MOTOR_R_INA (N1) or MOTOR_L_INA (N2), SDO to MOTOR_R_INB (N1) or MOTOR_L_INB (N2), EDP to MOTOR_R_SEL (N1) or MOTOR_L_SEL (N2), OUTA1 to MOTOR_R_OUTA (N1) or MOTOR_L_OUTA (N2), OUTA2 to MOTOR_R_OUTA (N1) or MOTOR_L_OUTA (N2), OUTB1 to MOTOR_R_OUTB (N1) or MOTOR_L_OUTB (N2), and OUTB2 to MOTOR_R_OUTB (N1) or MOTOR_L_OUTB (N2).



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Benennung:		Klasse: 5BHEL	Dokumentenart: Schaltplan
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		Version: 1.0	
		Dokumentstatus: Prototyp	
		Datum: 01.10.2025	Blatt: 1
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